Chapter 5

Function of Several Variables

Definition: A function f of two variables is a rule that assigns to each ordered pair of real numbers (x, y) in a set D a unique real number denoted by $f(x, y)$.

Partial Derivatives: If $z = f(x, y)$, the first partial derivatives of z,

$$f_x(x, y) = f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x,\ y) = \frac{\partial z}{\partial x} = f_1 = D_1 f = D_x f$$

$$f_y(x, y) = f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x,\ y) = \frac{\partial z}{\partial y} = f_2 = D_2 f = D_y f$$

- To find f_x, regard y as a constant and differentiate $f(x, y)$ with respect to x, and
- To find f_y, regard x as a constant and differentiate $f(x, y)$ with respect to y.

Example 1: If $f(x, y) = x^3 + x^2y^3 - 2y^2$, find $f_x(2, 1)$ and $f_y(2, 1)$.

Solution: Holding y constant and differentiating with respect to x, we get,

$$f_x(x, y) = 3x^2 + 2xy^3$$

and so,

$$f_x(2, 1) = 3 \times 2^2 + 2 \times 2 \,.\, 1^3 = 16$$

Holding x constant and differentiating with respect to y, we get

$$f_y(x, y) = 3x^2y^2 - 4y$$

and so,

$$f_y(2, 1) = 3 \times 2^2 \times 1^2 - 4 \times 1 = 8$$

Example 2: If $f(x, y) = \sin\left(\frac{x}{1+y}\right)$, calculate $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Solution: Using the Chain Rule for functions of one variable, we have,

$$\frac{\partial f}{\partial x} = \cos\left(\frac{x}{1+y}\right).\frac{\partial}{\partial x}\left(\frac{x}{1+y}\right) = \cos\left(\frac{x}{1+y}\right).\frac{1}{1+y}$$

and

$$\frac{\partial f}{\partial y} = \cos\left(\frac{x}{1+y}\right).\frac{\partial}{\partial y}\left(\frac{x}{1+y}\right) = -\cos\left(\frac{x}{1+y}\right).\frac{x}{(1+y)^2}$$

Problem Set 5.1

Find the first partial derivatives of the function.

1. $f(x,y) = x^4 + 5xy^3$
2. $f(x,y) = x^2y - 3y^4$
3. $f(x,t) = t^2e^{-x}$
4. $f(x,t) = \sqrt{3x+4t}$
5. $\mathrm{z} = \ln(x+t^2)$
6. $\mathrm{z} = x\sin(xy)$
7. $f(x,y) = y\sin^{-1}(xy),\ f_y\left(1,\frac{1}{2}\right)$
8. $R(s,t) = te^{\frac{s}{t}}\ ,\ R_t(0,1)$
9. $f(x,y) = \frac{x}{y}$
10. $f(x,y) = \frac{x}{(x+y)^2}$
11. $g(u,v) = (u^2v - v^3)^5$
12. $u(r,\theta) = \sin(r\cos\theta)$

[**Ref.** James Stewart 8th edition, page-924]

Higher Derivatives: If $\mathrm{z} = f(x,y)$, the second partial derivatives of z,

$$(f_x)_x = f_{xx} = f_{11} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 z}{\partial x^2} \qquad (f_y)_x = f_{yx} = f_{21} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 z}{\partial x \partial y}$$

$$(f_x)_y = f_{xy} = f_{12} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 z}{\partial y \partial x} \qquad (f_y)_y = f_{yy} = f_{22} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 z}{\partial y^2}$$

Example 3: Find the second partial derivatives of

$$f(x,y) = x^3 + x^2y^3 - 2y^2$$

Solution: We get first partial derivatives from **example 1,**

$$f_x(x,y) = 3x^2 + 2xy^3 \qquad f_y(x,y) = 3x^2y^2 - 4y$$

Therefore,

$$f_{xx} = \frac{\partial}{\partial x}(3x^2 + 2xy^3) = 6x + 2y^3 \qquad f_{yx} = \frac{\partial}{\partial x}(3x^2y^2 - 4y) = 6xy^2$$

$$f_{xy} = \frac{\partial}{\partial y}(3x^2 + 2xy^3) = 6xy^2 \qquad f_{yy} = \frac{\partial}{\partial y}(3x^2y^2 - 4y) = 6x^2y - 4$$

Example 4: Calculate f_{yxy}, if $f(x,y) = \sin(2x+5y)$.

Solution:

$$f_y = \cos(2x+5y)\cdot 5$$

$$f_{yx} = -5\sin(2x+5y)\cdot 2 = -10\sin(2x+5y)$$

$$f_{yxy} = -10\cos(2x+5y)\cdot 5 = -50\cos(2x+5y)$$

Problem Set 5.2

[1-5] Find the second partial derivatives of the function.

1. $f(x,y) = x^4y - 2y^2$
2. $f(x,y) = \ln(ax + by)$
3. $z(x,y) = \frac{y}{2x+3y}$
4. $T(r,\theta) = e^{-2r}\cos\theta$
5. $v(s,t) = \sin(s^2 - t^2)$

[6-9] Verify that the conclusion of **Clairaut's Theorem** holds, that is, $\boldsymbol{u_{xy} = u_{yx}}$.

6. $u(x,y) = x^4y^3 - y^4$
7. $u(x,y) = e^{xy}\sin y$
8. $u(x,y) = \cos(x^2y)$
9. $u(x,y) = \ln(x + 2y)$

[10-15] Determine whether the following function is a solution of **Laplace equations**, $\boldsymbol{u_{xx} + u_{yy} = 0}$.

10. $u(x,y) = x^2 + y^2$
11. $u(x,y) = \ln\sqrt{x^2 + y^2}$
12. $u(x,y) = \sin x\cosh y + \cos x\sinh y$
13. $u(x,y) = e^x\sin y$
14. $u(x,y) = e^{-x}\cos y - e^{-y}\cos x$
15. $u(x,y) = \sin x\cosh y + \cos x\sinh y$

[16-19] Determine whether the following function is a solution of the **Wave equation**, $\boldsymbol{u_{tt} = a^2u_{xx}}$.

16. $u(x,t) = \sin(x - at)$
17. $u(x,t) = (x - at)^6 + (x + at)^6$
18. $u(x,t) = \sin(x - at) + \ln(x + at)$
19. $u(x,t) = \sin(kx)\sin(akt)$

[**Ref.** James Stewart 8th edition, page-925]

Taylor polynomial for multiple variable functions:

For a function of two variables $f(x,y)$ whose first and second partial derivatives exist at the point (a,b), the **2nd-degree Taylor polynomial** of $f(x,y)$ near the point (a,b) is:

$$f(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b) + \frac{f_{xx}(a,b)}{2}(x-a)^2 + f_{xy}(a,b)(x-a)(y-b) + \frac{f_{yy}(a,b)}{2}(y-b)^2$$

Example: Determine the 1st- and 2nd-degree Taylor polynomial approximations for the following function $f(x,y)$ of x and y near the given point.

$f(x,y) = \sin 2x + \cos y$ for (x,y) near the point $(0,0)$.

Solution:

$f_x(x,y) = 2\cos 2x$ and $f_y(x,y) = -\sin y$

$f_{xx}(x,y) = -4\sin 2x$, $f_{xy}(x,y) = 0$ and $f_{yy}(x,y) = -\cos y$

At point $(0,0)$,

$$f(0,0) = \sin 2(0) + \cos 0 = 1 \qquad f_{xx}(0,0) = -4\sin 2(0) = 0$$

$$f_x(0,0) = 2\cos 2(0) = 2 \qquad f_{xy}(0,0) = 0$$

$$f_y(0,0) = -\sin 0 = 0 \qquad f_{yy}(0,0) = -\cos 0 = -1$$

∴ 2nd-**degree** Taylor polynomial approximations is,

$$f(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b) + \frac{f_{xx}(a,b)}{2}(x-a)^2 + f_{xy}(a,b)(x-a)(y-b) + \frac{f_{yy}(a,b)}{2}(y-b)^2$$

$$f(x,y) = f(0,0) + f_x(0,0)(x-0) + f_y(0,0)(y-0) + \frac{f_{xx}(0,0)}{2}(x-0)^2 + f_{xy}(0,0)(x-0)(y-0) + \frac{f_{yy}(0,0)}{2}(y-0)^2$$

$$= 1 + 2x + \frac{0}{2}x^2 + (0)xy + \frac{-1}{2}y^2 == 1 + 2x - \frac{y^2}{2}.$$

Problem Set 5.3

Determine the 1st and 2nd-degree Taylor polynomial approximations for the following functions $f(x,y)$ of x and y near the given point.

(i) $f(x,y) = xe^y + 1$ for (x,y) near the point $(1,0)$.

(ii) $f(x,y) = x\sqrt{y}$ for (x,y) near the point $(1,2)$.

(iii) $f(x,y) = e^x \cos y$ for (x,y) near the point $(0,0)$.

Chapter 6

Maximum and Minimum Values

Definition:

A function of two variables $f(x,\ y)$ has a

- **local maximum** at (a,b), if $f(x,y) \leq f(a,b)$
- **local minimum** at (a,b), if $f(x,y) \geq f(a,b)$

when (x,y) is near (a,b).

Then $f(a,b)$ is a **local maximum value** or **local minimum value.**

Stationary point or critical point:

A point (a,b) is called a Critical point (or stationary point) of $f(x,y)$ if

$f_x(a,b) = 0$ and $f_y(a,b) = 0$

Second Derivatives Test:

Let (a,b) be a critical point of $f(x,\ y)$ and

$$D(a,b) = f_{xx}(a,b)f_{yy}(a,b) - \left[\, f_{xy}(a,\ b)\right]^2$$

- If $D > 0$ and $f_{xx}(a,b) > 0$, then $f(a,b)$ is a local minimum.
- If $D > 0$ and $f_{xx}(a,b) < 0$, then $f(a,b)$ is a local maximum.
- If $D < 0$, then $f(a,b)$ is not a local maximum or minimum, f has a saddle point at (a,b).
- If $D = 0$, then no conclusion can be drawn.

Example: Find the local maximum and minimum values and saddle points of $f(x,y) = x^4 + y^4 - 4xy + 1$.

Solution: To locate the critical points setting partial derivatives equal to 0,

$f_x = 4x^3 - 4y = 0$

$\Rightarrow x^3 - y = 0$

$\Rightarrow x^3 = y \qquad (1)$

$f_y = 4y^3 - 4x = 0$

$\Rightarrow y^3 - x = 0$

$\Rightarrow y^3 = x \qquad (2)$

To solve these equations, we substitute $y = x^3$ from the equation (1) into the equation (2). We get,

$$x^9 - x = 0 \Rightarrow x(x^8 - 1) = 0 \Rightarrow x = 0, or\ x^8 = 1 \Rightarrow x = \pm 1$$

The three critical points are $(0,0)$, $(-1,-1)$ and $(1,1)$.

Next, we calculate the second partial derivatives and $D(x,y)$.

$f_{xx} = 12x^2$ $\qquad$ $f_{xy} = -4$ $\qquad$ $f_{yy} = 12y^2$

$$D(x,y) = f_{xx}f_{yy} - \left(f_{xy}\right)^2 = 144x^2y^2 - 16$$

Stationary points	$D(a,b)$	$f_{xx}(a,b)$	**Results**
$(0,\ 0)$	-16		saddle point
$(-1,-1)$	128	12	local minimum
$(1,\ 1)$	128	12	local minimum

Therefore, local minimum $= f(-1,-1) = (-1)^4 + (-1)^4 - 4(-1)(-1) + 1 = -1$.

Problem Set 6

Find the local maximum and minimum values and saddle point(s) of the functions.

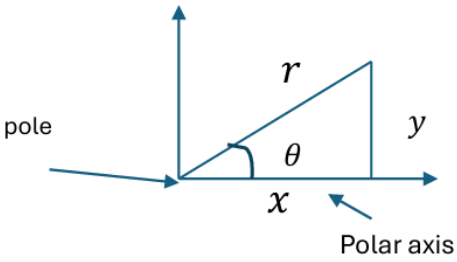
1. $f(x,y) = x^2 + xy + y^2 + y$
2. $f(x,y) = xy - 2x - 2y - x^2 - y^2$
3. $f(x,y) = x^2 + y^4 + 2xy$
4. $f(x,y) = 2 - x^4 + 2x^2 - y^2$
5. $f(x,y) = x^3 - 3x + 3xy^2$
6. $f(x,y) = x^3 + y^3 - 3x^2 - 3y^2 - 9x$
7. $f(x,y) = x^4 - 2x^2 + y^3 - 3y$ [1-7] [**Ref.** James Stewart 8th edition, page-968]

Chapter 7

Coordinate Geometry of Two Dimensions

Polar Coordinates

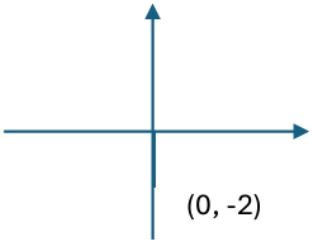
- Pole
- Polar axis
- Relation between Polar and Cartesian



$x = r\cos\theta, \quad y = r\sin\theta \text{ and } r^2 = x^2 + y^2, \quad \tan\theta = \frac{y}{x}$

Example 1: Plot the point and find the Cartesian coordinate of the point $\left(2, \frac{3\pi}{2}\right)$.

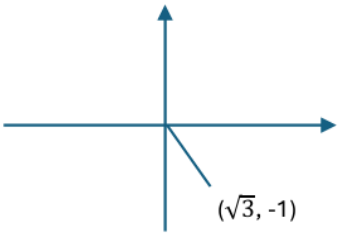
Given, $r = 2,\ \theta = \frac{3\pi}{2}$ Now $x = 2\cos\frac{3\pi}{2} = 0,\ y = 2\sin\frac{3\pi}{2} = -2$



Example 2: The Cartesian coordinate of a point is given, find polar coordinates (r, θ).

Given $\left(\sqrt{3}, -1\right),\ x = \sqrt{3}, y = -1 \quad \therefore r = \sqrt{x^2 + y^2} = \sqrt{3} + 1 = 2$

$$\tan\theta = \frac{y}{x} = -\frac{1}{\sqrt{3}} = -\tan\frac{\pi}{6} = \tan\left(2\pi - \frac{\pi}{6}\right) \quad \therefore \theta = \frac{11\pi}{6}$$



Problem Set 7.1

1. Plot the point whose polar coordinates are given. Then find the Cartesian coordinates of the points.

(a) $(2, 3\pi/2)$ (b) $(4, 4\pi/3)$ (c) $(-1, -\pi/6)$ (d) $(-2, 3\pi/4)$ (e) $(-3, -\pi/3)$

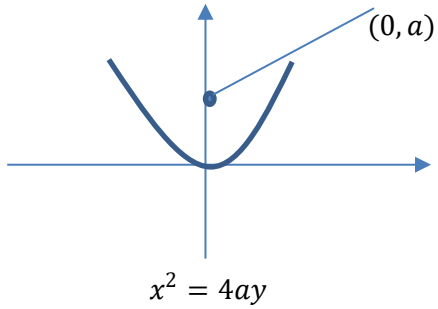
2. Find the polar coordinates (r, θ) from the Cartesian coordinates.

(a) $(-4, 4)$ (b) $(3, 3\sqrt{3})$ (c) $(\sqrt{3}, -1)$ (d) $(-6, 0)$ (e) $(0, -7)$

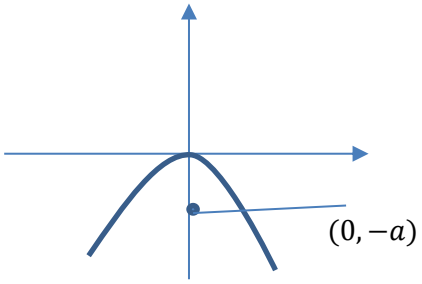
Parabola: Equation of Parabola:

Equation	Vertex	Axis along	Distance of the focus from the vertex
$x^2 = 4ay$	$(0, 0)$	y-axis	a
$y^2 = 4ax$	$(0, 0)$	x-axis	a
$(x-h)^2 = 4a(y-k)$	(h, k)	y-axis	a
$(y-k)^2 = 4a(x-h)$	(h, k)	x-axis	a

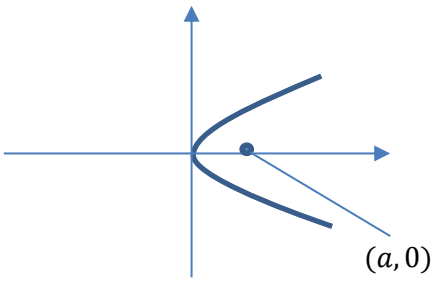
- Graphs of Parabola:



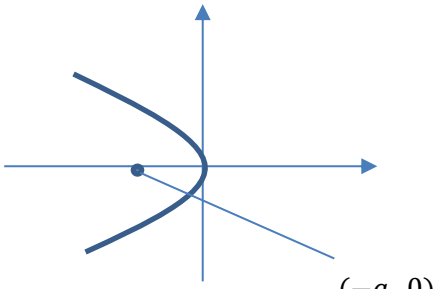
$x^2 = 4ay$



$x^2 = -4ay$



$y^2 = 4ax$



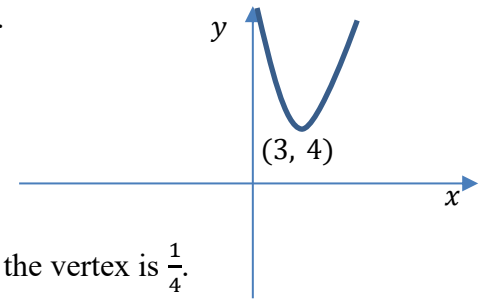
$y^2 = -4ax$

Example 1: Identify the type of curve and sketch the graph, $y = x^2 - 6x + 13$.

Solution: To reduce in general equation,

$y = x^2 - 2(x)(3) + 3^2 - 9 + 13 = (x-3)^2 + 4$

We get, $(x-3)^2 = 4.\frac{1}{4}(y-4)$

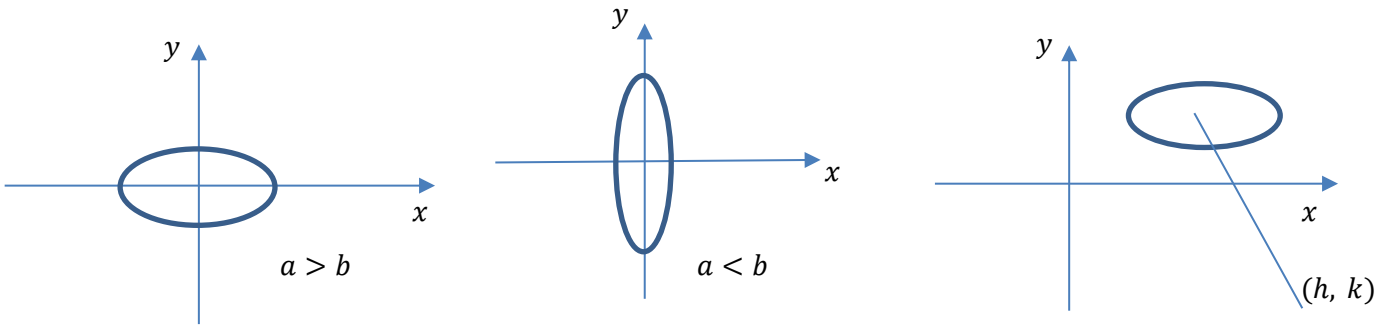


This represents a parabola with vertex (3, 4) and the distance of the focus from the vertex is $\frac{1}{4}$.

Ellipse: Equation of Ellipse:

Equation	Dist. Of the focus from the Centre	Centre	Vertex	Major axis along	Minor axis
$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, a>b$	$c=\sqrt{a^2-b^2}$	(0, 0)	$(\pm a,\ 0)$	X-axis	Y-axis
$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, a<b$	$c=\sqrt{b^2-a^2}$	(0, 0)	$(0,\ \pm b)$	Y-axis	X-axis
$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$		(h, k)			

- Graphs of Ellipse:



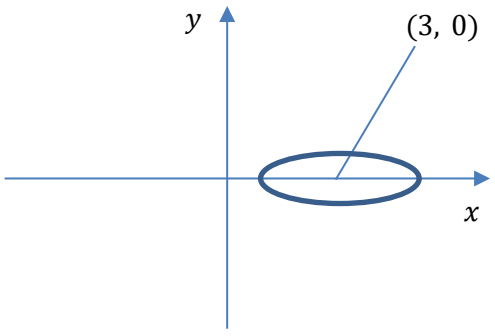
Example 2: Identify the type of curve and sketch the graph, $x^2+4y^2-6x+5=0$.

Solution: To reduce in general equation,

$$x^2-2(x)(3)+3^2-9+4y^2+5=0$$

We get, $(x-3)^2+4y^2=4 \quad \rightarrow \quad \frac{(x-3)^2}{4}+\frac{y^2}{1}=1.$

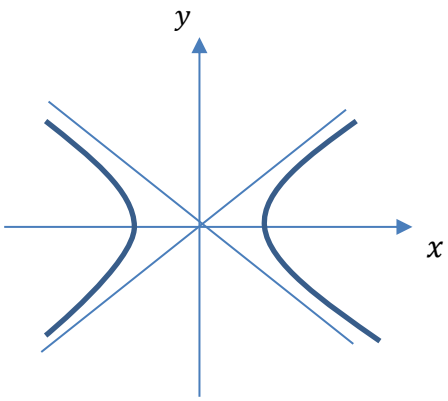
This represents an **ellipse** with center $(3,0)$, vertices $(1,0)$ & $(5,0)$ and the distance of the foci from the center $c=\sqrt{4-1}=\sqrt{3}$.



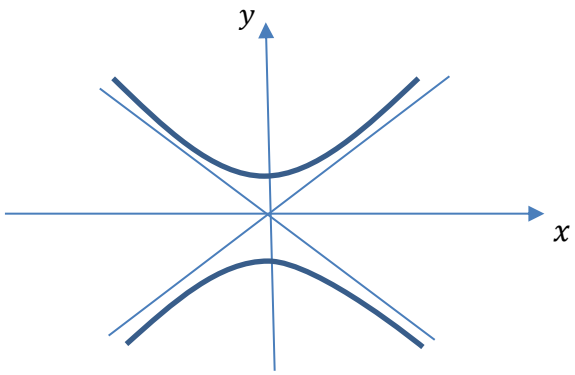
Hyperbola: Equation of Hyperbola,

Equation	**Distance of the focus from the Centre**	**Centre**	**Vertex**	**Asymptotes**
$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ (Transverse axis is along x-axis)	$c=\sqrt{a^2+b^2}$	(0, 0)	$(\pm a,\ 0)$	$y=\pm\frac{b}{a}x$
$\frac{y^2}{b^2}-\frac{x^2}{a^2}=1$ (Transverse axis is along y-axis)	$c=\sqrt{a^2+b^2}$	(0, 0)	$(0,\ \pm b)$	$y=\pm\frac{b}{a}x$
$\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$	$c=\sqrt{a^2+b^2}$	(h, k)		$y=k\pm\frac{b}{a}(x-h)$

- Graphs of Hyperbola:



$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$



$\frac{y^2}{b^2}-\frac{x^2}{a^2}=1$

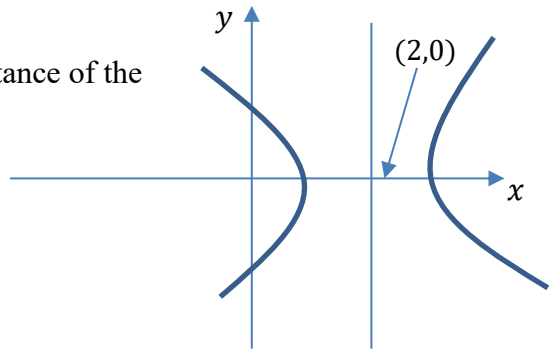
Example 3: Identify the type of curve and sketch the graph, $x^2-y^2-4x+3=0$.

Solution: To reduce in general equation,

$$x^2-2(x)(2)+2^2-4-y^2+3=0.$$

We get,$(x-2)^2-y^2=1$

This represents a **hyperbola** with center (2, 0), the distance of the foci from the center $c=\sqrt{1+1}=\sqrt{2}$ and the asymptotes $y=\pm(x-2)$.



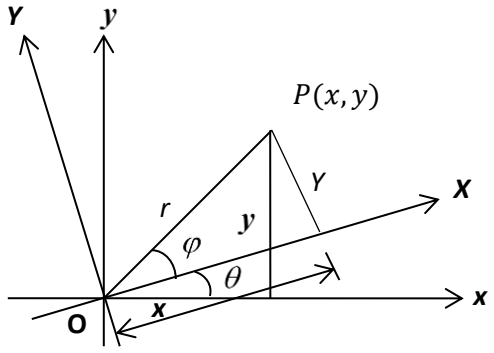
Problem Set 7.2

Identify the type of curve and sketch the graph.

1. $y = -x^2$
2. $y^2 - x^2 = 1$
3. $x^2 + 4y^2 = 16$
4. $x = -2y^2$
5. $16x^2 - 25y^2 = 400$
6. $25x^2 + 4y^2 = 100$
7. $4x^2 + y^2 = 1$
8. $y = x^2 + 2$
9. $x = y^2 - 1$
10. $9x^2 - 25y^2 = 225$
11. $9y^2 - x^2 = 9$
12. $2x^2 + 5y^2 = 10$
13. $y = x^2 + 2x$
14. $9(x-1)^2 + 4(y-2)^2 = 36$
15. $16x^2 + 9y^2 - 36y = 108$
16. $x = 4 - y^2$
17. $y^2 - 2x + 6y + 5 = 0$
18. $4x^2 + 9y^2 - 16x + 54y + 61 = 0$

[**Ref.** James Stewart 8th edition, page-A23]

Rotation of Axes



Let, $P(x, y)$ be a point under xy −coordinate system. $OP = r$ makes an angle ϕ with positive x −axis.

$x = r\cos\phi, \quad y = r\sin\phi.$

Suppose that the coordinate system is rotated through an angle θ keeping the origin fixed. If P has coordinates (X, Y) referred to the new axes, then

$$X = r\cos(\phi - \theta)$$
$$\Rightarrow X = r\cos\phi\cos\theta + r\sin\phi\sin\theta$$
$$\Rightarrow X = x\cos\theta + y\sin\theta \ldots\ldots\ldots (1)$$

$$Y = r\sin(\phi - \theta)$$
$$\Rightarrow Y = r\sin\phi\cos\theta - r\cos\phi\sin\theta$$
$$\Rightarrow Y = y\cos\theta - x\sin\theta.$$
$$\Rightarrow Y = -x\sin\theta + y\cos\theta \ldots\ldots\ldots\ldots (2)$$

Converting equation (1) and equation (2) into matrix form we get,

$$\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \qquad \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$\Rightarrow x = X\cos\theta - Y\sin\theta$$
$$y = X\sin\theta + Y\cos\theta.$$

General Equation of Second Degree

The general equation of second degree is

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ (1)

Where a, h, b, g, f and c are constant.

Angle of rotation to remove xy term from general equation of second degree

The general equation of second degree is

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ (1)

Under rotation through an angle θ, the transformation equations are

$$x = X\cos\theta - Y\sin\theta$$

$$y = X\sin\theta + Y\cos\theta$$

and the general equation of second degree becomes

$$a(X\cos\theta - Y\sin\theta)^2 + 2h(X\cos\theta - Y\sin\theta)(X\sin\theta + Y\cos\theta)$$

$$b(X\sin\theta + Y\cos\theta)^2 + 2g(X\cos\theta - Y\sin\theta) + 2f(X\cos\theta - Y\sin\theta) + c = 0$$

which can be written as

$$AX^2 + 2HXY + BY^2 + 2GX + 2FY + C = 0$$

Where, $A = a\cos^2\theta + 2h\cos\theta\sin\theta + b\sin^2\theta$

$$B = a\sin^2\theta - 2h\cos\theta\sin\theta + b\cos^2\theta$$

$$H = h(\cos^2\theta - \sin^2\theta) - (a-b)\sin\theta\cos\theta$$

$$G = g\cos\theta + f\sin\theta$$

$$F = -g\sin\theta + f\cos\theta$$

$$C = c$$

The transformed equation will be independent of the product term XY if $H = 0$ i.e.

$$h(\cos^2\theta - \sin^2\theta) - (a-b)\sin\theta\cos\theta = 0$$

$$h\cos 2\theta = \frac{1}{2}(a-b)\sin 2\theta$$

$$\tan 2\theta = \frac{2h}{a-b}.$$

The nature of the **conics** represent by (1)

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

may be predicted by the sign of the discriminant

$$\Delta = h^2 - ab.$$

Results are summarized below:

Conditions on $\Delta = h^2 - ab$	**Regular Conics**
$\Delta < 0$	Ellipse
$\Delta > 0$	Hyperbola
$\Delta = 0$	Parabola

To reduce standard form, we may follow the following steps:

Rotate the coordinate axes through an angle θ such that $\cot 2\theta = \frac{a-b}{2h}$ to get rid of xy term and then use translation to reduce to standard form.

Example 4: Find the angle of rotation to remove xy term from $5x^2 + 4xy + 2y^2 = 6$ and hence write the transformed equation. Also reduce them to standard form and sketch them showing both set of axes.

Solution: Identifying with $a = 5,\ \ b = 2$ and $2h = 4$ and choosing θ such that

$$cot\, 2\,\theta = \frac{a-b}{2h} = \frac{5-2}{4} = \frac{3}{4}$$

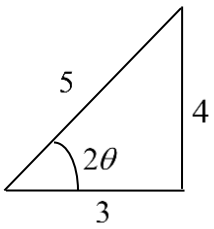
From the triangle we see that

$$cos\, 2\,\theta = \frac{3}{5} \Rightarrow \theta = \frac{1}{2}\text{arc}\, cos\left(\frac{3}{5}\right) = 26.5^0$$

The values of $cos\,\theta$ and $sin\,\theta$ can then be computed from the half-angle formulas:

$$\cos\theta = \sqrt{\frac{1+cos\,2\theta}{2}} = \sqrt{\frac{1+\frac{3}{5}}{2}} = \frac{2}{\sqrt{5}}$$

$$\sin\theta = \sqrt{\frac{1-cos\,2\theta}{2}} = \sqrt{\frac{1-\frac{3}{5}}{2}} = \frac{1}{\sqrt{5}}$$



The transformation equations become

$$x = X\cos\theta - Y\sin\theta = \frac{1}{\sqrt{5}}(2X - Y)$$

And $y = X\sin\theta + Y\cos\theta = \frac{1}{\sqrt{5}}(X + 2Y)$

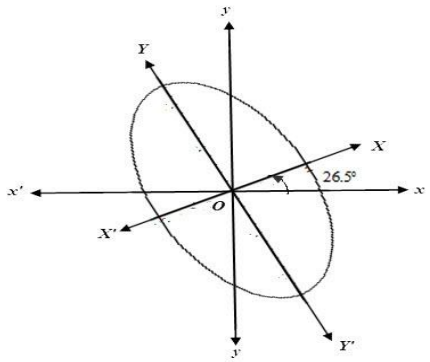
Substitution in the original equation gives

$$\frac{5}{5}(2X - Y)^2 + \frac{4}{5}(2X - Y)(X + 2Y) + \frac{2}{5}(X + 2Y)^2 = 6$$

Collecting similar terms

$$\frac{1}{5}(20 + 8 + 2)X^2 + \frac{1}{5}(-20 + 16 - 4 + 8)XY + \frac{1}{5}(5 - 8 + 8)Y^2 = 6$$

or $\quad 6X^2 + Y^2 = 6. \Rightarrow \frac{X^2}{1^2} + \frac{Y^2}{(\sqrt{6})^2} = 1.$



Example 5: Find the angle of rotation to remove xy term from $11x^2 + 24xy + 4y^2 - 12x + 16y = 0$ and then write the transformed equation. Also reduce them to standard form and sketch them showing both set of axes.

Solution: Identifying with $a = 11,\ \ b = 4$ and $2h = 24$and choosing θ such that

$$\cot 2\theta = \frac{a-b}{2h} = \frac{11-4}{24} = \frac{7}{24}$$

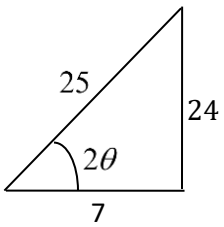
From the triangle we see that

$$\cos 2\theta = \frac{7}{25} \Rightarrow \theta = \frac{1}{2}\arccos\left(\frac{7}{25}\right) \approx 37°$$

The values of $\cos\theta$ and $\sin\theta$ can then be computed from the half-angle formulas:

$$cos\,\theta = \sqrt{\frac{1+cos\,2\theta}{2}} = \sqrt{\frac{1+\frac{7}{25}}{2}} = \frac{4}{5}$$

$$sin\,\theta = \sqrt{\frac{1-cos\,2\theta}{2}} = \sqrt{\frac{1-\frac{7}{25}}{2}} = \frac{3}{5}$$



The transformation equations become

$$x = X\,cos\,\theta - Y\,sin\,\theta = \frac{1}{5}(4X - 3Y)$$

And $y = X\,sin\,\theta + Y\,cos\,\theta = \frac{1}{5}(3X + 4Y)$

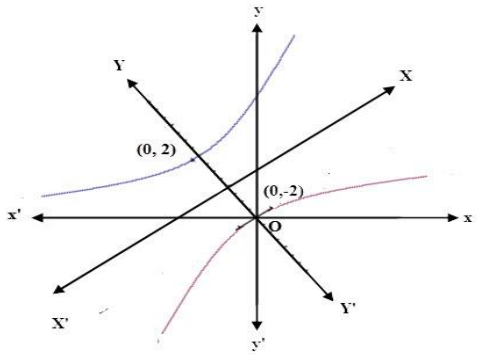
Substitution in the original equation give

$$\frac{11}{25}(4X-3Y)^2 + \frac{24}{25}(4X-3Y)(3X+4Y) + \frac{4}{25}(3X+4Y)^2 - \frac{12}{5}(4X-3Y) + \frac{16}{5}(3X+4Y) = 0$$

Collecting similar terms

$$\frac{1}{25}(176 + 288 + 36)X^2 + \frac{1}{25}(99 - 288 + 64)Y^2 + \frac{1}{5}(36+64)Y = 0$$

or $\qquad 20X^2 - 5Y^2 + 20Y = 0 \qquad \Rightarrow \frac{(Y-2)^2}{2^2} - \frac{X^2}{1^2} = 1\,.$



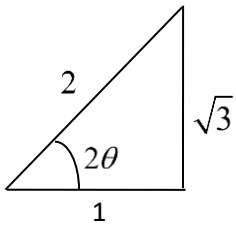
Example 6: Find the angle of rotation to remove xy term from $x^2 - 2\sqrt{3}xy + 3y^2 - 16\sqrt{3}x - 16y = 0$ and hence write the transformed equation. Also reduce them to standard form and sketch them showing both set of axes.

Solution: Identifying with $a = 1,\quad b = 3$ and $2h = -2\sqrt{3}$ and choosing θ such that

$$cot\, 2\,\theta = \frac{a-b}{2h} = \frac{1-3}{-2\sqrt{3}} = \frac{1}{\sqrt{3}}$$

From the triangle we see that

$$cos\, 2\,\theta = \frac{1}{2} \Rightarrow 2\theta = \frac{\pi}{3} \Rightarrow \theta = \frac{\pi}{6}$$



The transform equations are

$$x = X\, cos\, \theta - Y\, sin\, \theta = \frac{1}{2}(\sqrt{3}X - Y)$$

and$y = X\, sin\, \theta + Y\, cos\, \theta = \frac{1}{2}(X + \sqrt{3}Y)$

Substitution in the original equation gives

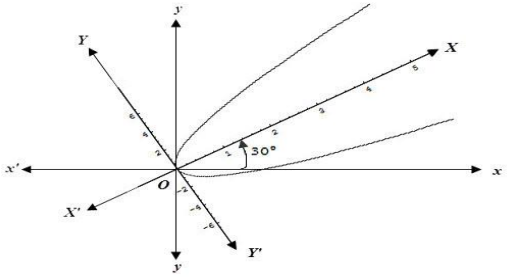
$$\frac{1}{4}(\sqrt{3}X - Y)^2 - \frac{2\sqrt{3}}{4}(\sqrt{3}X - Y)(X + \sqrt{3}Y) + \frac{3}{4}(X + \sqrt{3}Y)^2 - \frac{16\sqrt{3}}{2}(\sqrt{3}X - Y) - \frac{16}{2}(X + \sqrt{3}Y) = 0$$

Collecting similar terms

$$\frac{1}{4}(3 - 6 + 3)X^2 + \frac{1}{4}(-2\sqrt{3} - 6\sqrt{3} + 2\sqrt{3} + 6\sqrt{3})XY + \frac{1}{4}(1 + 6 + 9)Y^2 - 32X = 0$$

or $4X^2 = 32X$

or $Y^2 = 8X$.



Problem Set 7.3

1. Find the angle of rotation to remove *xy*-term from the following curves.

   (a) $4x^2 + 8xy - 4y^2 - 3x = 5$

   (b) $3\sqrt{3}x^2 + 2xy + \sqrt{3}y^2 = 16$

   (c) $x^2 + \sqrt{3}xy + y^2 = 6.$

2. The Coordinate axes are rotated by the following given angle θ. Find the transformed equations of the following curves. Also reduce them to standard form and sketch them showing both set of axes.

   (a) $7x^2 - 6\sqrt{3}xy + 13y^2 = 64,\ \theta = \frac{\pi}{6}$

   (b) $5x^2 - 8xy + 5y^2 = 9,\ \theta = \frac{\pi}{4}$

   (c) $4x^2 - 10xy + 4y^2 + 9 = 0,\ \theta = \frac{\pi}{4}$

   (d) $x^2 - 10\sqrt{3}xy + 11y^2 + 4 = 0,\ \theta = \frac{\pi}{6}$

   (e) $x^2 - 2xy + y^2 + 4\sqrt{2}x + 4\sqrt{2}y - 16 = 0,\ \theta = \frac{\pi}{4}$

   (f) $x^2 - 2\sqrt{3}xy + 3y^2 - 16\sqrt{3}x - 16y = 0, \theta = \frac{\pi}{6}.$

Chapter 8

Three-Dimensional Coordinate Geometry

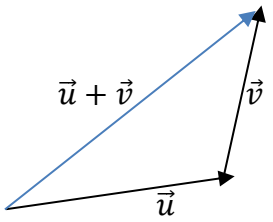
Vector: In mathematics and physics, a **vector** is an object that has both **magnitude** (length or size) and **direction**. Vectors are typically represented as arrows in space, where the length of the arrow corresponds to the magnitude, and the arrow points in the direction of the vector.

A **vector** is defined by two points:

- **Initial Point (Tail):** The starting location of the vector.
- **Terminal Point (Head):** The ending location of the vector.

Vector Addition: **Vector addition** is the operation of combining two or more vectors to produce a new vector called the **resultant vector**. This operation follows specific geometric and algebraic rules.

Triangle Law of Vector Addition: If $\vec{u}$ and $\vec{v}$ are vectors positioned so the initial point of $\vec{v}$ is the terminal point of $\vec{u}$, then the sum $\vec{u}+\vec{v}$ is the vector from the initial point of $\vec{u}$ to the terminal point of $\vec{v}$.



Scalar Multiplication: If c is a scalar and $\vec{v}$ is a vector, then the scalar multiple $c\vec{v}$ is the vector whose length is $|c|$ times the length of $\vec{v}$ and whose direction is the same as $\vec{v}$ if $c>0$ and is opposite to $\vec{v}$ if $c<0$. If $c=0$ or $\vec{v}=\vec{0}$, then $c\vec{v}=\vec{0}$.

Position Vector: Given the points $A(x_1,y_1,z_1)$ and $B(x_2,y_2,z_2)$, the vector $\overrightarrow{AB}$ is

$$\overrightarrow{AB}=\langle x_2-x_1,y_2-y_1,z_2-z_1\rangle.$$

Magnitude: The magnitude or length of the two-dimensional vector $\vec{a}=\langle a_1,a_2\rangle$ is

$$|\vec{a}|=\sqrt{a_1^2+a_2^2}$$

The magnitude or length of the three-dimensional vector $\vec{a}=\langle a_1,a_2,a_3\rangle$ is

$$|\vec{a}|=\sqrt{a_1^2+a_2^2+a_3^2}.$$

Properties of Vectors: If $\vec{a},\vec{b},and\ \vec{c}$ are vectors in V_n and $c\ and\ d$ are scalars, then

1. $\vec{a}+\vec{b}=\vec{b}+\vec{a}$
2. $\vec{a}+(\vec{b}+\vec{c})=(\vec{a}+\vec{b})+\vec{c}$
3. $\vec{a}+\vec{0}=\vec{a}$
4. $\vec{a}+\overrightarrow{(-a)}=\vec{0}$

5. $c(\vec{a}+\vec{b}) = c\vec{a} + c\vec{b}$
6. $(c+d)\vec{a} = c\vec{a} + d\vec{a}$
7. $(cd)\vec{a} = c(d\vec{a})$
8. $1\vec{a} = \vec{a}$

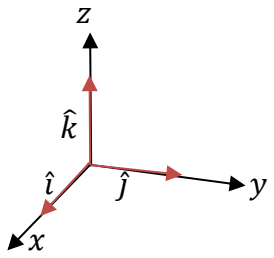
Standard Basis Vectors: The vectors $\hat{\imath}, \hat{\jmath}$, and $\hat{k}$ are called standard basis vectors where

$$\hat{\imath} = \langle 1,0,0\rangle \quad \hat{\jmath} = \langle 0,1,0\rangle \quad \hat{k} = \langle 0,0,1\rangle$$

The magnitude of these vectors is 1, that is,

$$|\hat{\imath}| = 1 \quad |\hat{\jmath}| = 1 \quad |\hat{k}| = 1$$

The directions of $\hat{\imath}, \hat{\jmath}$, and $\hat{k}$ are along positive x, y, and z-axes respectively.



Unit Vector: A unit vector is a vector whose magnitude is 1. For instance, $\hat{\imath}, \hat{\jmath}$, and $\hat{k}$ are all unit vectors. In general, if $|\vec{a}| \neq 0$, then the unit vector that has the same direction as $\vec{a}$ is,

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|}$$

Dot Product: If $\vec{a} = \langle a_1, a_2, a_3\rangle$ and $\vec{b} = \langle b_1, b_2, b_3\rangle$, then the dot product of $\vec{a}$ and $\vec{b}$ is the number $\vec{a}\cdot\vec{b}$ given by

$$\vec{a}\cdot\vec{b} = a_1b_1 + a_2b_2 + a_3b_3$$

Theorem: If θ is the angle between the vectors $\vec{a}$ and $\vec{b}$, then

$$\vec{a}\cdot\vec{b} = |\vec{a}|\,|\vec{b}|\cos\theta$$

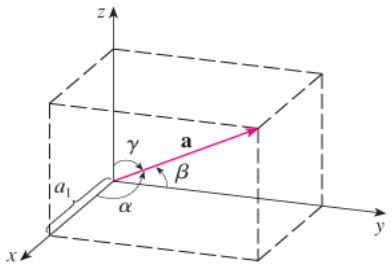
$$\text{Therefore, } \cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}|\,|\vec{b}|}$$

- Two vectors $\vec{a}$ and $\vec{b}$ are perpendicular or orthogonal if and only if $\vec{a}\cdot\vec{b} = 0$.

Direction Angles and Direction Cosines: The direction angles of a nonzero vector $\vec{a}$ are the angles α, β, and γ that $\vec{a}$ makes with the positive x, y and z-axes, respectively. $\cos\alpha$, $\cos\beta$, and $\cos\gamma$ are called the direction cosines of the vector $\vec{a}$. Let, $\vec{a} = \langle a_1, a_2, a_3\rangle$, then

$$\cos\alpha = \frac{\vec{a}\cdot\hat{\imath}}{|\vec{a}||\hat{\imath}|} = \frac{a_1}{|\vec{a}|}$$

Similarly, $\cos\beta = \frac{a_2}{|\vec{a}|}$ and $\cos\gamma = \frac{a_3}{|\vec{a}|}$.



Scalar Projection: The scalar projection of $\vec{b}$ onto $\vec{a}$ is defined by $|\vec{b}|\cos\theta$, where θ is the angle between $\vec{a}$ and $\vec{b}$. This is denoted by $\text{comp}_a\vec{b}$. We know,

$$\vec{a}\cdot\vec{b} = |\vec{a}|\,|\vec{b}|\cos\theta$$

So, $|\vec{b}|\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}|}$

Therefore, scalar projection of $\vec{b}$ onto $\vec{a}$ is, $\text{comp}_a\vec{b} = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}|}$.

Similarly, scalar projection of $\vec{a}$ onto $\vec{b}$ is, $\text{comp}_b\vec{a} = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$.

Work Done: The work done W by the force $\vec{F}$ is defined to be the dot product $\vec{F}\cdot\vec{D}$, where $\vec{D}$ is the displacement vector.

$$W = \vec{F}\cdot\vec{D} = |\vec{F}||\vec{D}|\cos\theta$$

Example: If $\vec{a} = \langle 4,0,3\rangle$ and $\vec{b} = \langle -2,1,5\rangle$, find $\vec{a}\cdot\vec{b}$.

Solution: $\vec{a}\cdot\vec{b} = \langle 4,0,3\rangle\cdot\langle -2,1,5\rangle$

$= 4(-2) + 0\cdot 1 + 3\cdot 5$

$= -8 + 15 = 7$

Example: Find the angle between the vectors $\vec{a} = \langle 2,2,-1\rangle$ and $\vec{b} = \langle 5,-3,2\rangle$.

Solution: $|\vec{a}| = \sqrt{2^2+2^2+(-1)^2} = 3$

$|\vec{b}| = \sqrt{5^2+(-3)^2+2^2} = \sqrt{38}$

$\vec{a}\cdot\vec{b} = 2\cdot 5 + 2\cdot(-3) + (-1).2 = 2$

Therefore, $\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}|\,|\vec{b}|} = \frac{2}{3\sqrt{38}}$

So the angle between $\vec{a}$ and $\vec{b}$ is, $\theta = \cos^{-1}\left(\frac{2}{3\sqrt{38}}\right) \approx 84°$

Example: Show that $2\hat{\imath} + 2\hat{\jmath} - \hat{k}$ is orthogonal to $5\hat{\imath} - 4\hat{\jmath} + 2\hat{k}$.

Solution: $\left(2\hat{\imath} + 2\hat{\jmath} - \hat{k}\right)\cdot(5\hat{\imath} - 4\hat{\jmath} + 2\hat{k}) = 2\cdot 5 + 2\cdot(-4) + (-1)\cdot 2 = 0$

Since the dot product between the vectors is zero, they are orthogonal.

Example: Find the direction cosines and direction angles of the vector $\vec{a} = \langle 1,2,3\rangle$.

Solution: Let, $\alpha, \beta,$ and γ are direction angles and $\cos\alpha$, $\cos\beta$, and $\cos\gamma$ are direction cosines.

Here, $|\vec{a}| = \sqrt{1^2+2^2+3^2} = \sqrt{14}$.

$$\cos\alpha = \frac{\vec{a}\cdot\hat{\imath}}{|\vec{a}||\hat{\imath}|} \qquad \cos\beta = \frac{\vec{a}\cdot\hat{\jmath}}{|\vec{a}||\hat{\jmath}|} \qquad \cos\gamma = \frac{\vec{a}\cdot\hat{k}}{|\vec{a}||\hat{k}|}$$

So, the direction cosines are

$$\cos\alpha = \frac{1}{\sqrt{14}} \qquad \cos\beta = \frac{2}{\sqrt{14}} \qquad \cos\gamma = \frac{3}{\sqrt{14}}$$

and direction angles are

$$\alpha = \cos^{-1}\left(\frac{1}{\sqrt{14}}\right) \qquad \beta = \cos^{-1}\left(\frac{2}{\sqrt{14}}\right) \qquad \gamma = \cos^{-1}\left(\frac{3}{\sqrt{14}}\right).$$

Example: Find the scalar projection of $\vec{b} = \langle 1,1,2\rangle$ onto $\vec{a} = \langle -2,3,1\rangle$.

Solution: Here, $|\vec{a}| = \sqrt{(-2)^2 + 3^2 + 1^2} = \sqrt{14}$.

The scalar projection of $\vec{b}$ onto $\vec{a}$ is

$\text{comp}_a\vec{b} = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}|} = \frac{1(-2)+1\cdot 3+2\cdot 1}{\sqrt{14}} = \frac{3}{\sqrt{14}}$

Example: A force is given by the vector $\vec{F} = 3\hat{\imath} + 4\hat{\jmath} + 5\hat{k}$ and moves a particle from the point $P(2,1,0)$ to the point $Q(4,6,2)$. Find the work done.

Solution: The displacement vector is $\vec{D} = \overrightarrow{PQ} = Q - P$

$$= (4,6,2) - (2,1,0) = \langle 2,5,2\rangle$$

So the work done is

$W = \vec{F}\cdot\vec{D} = \langle 3,4,5\rangle \cdot \langle 2,5,2\rangle$

$= 6 + 20 + 10 = 36$

If the unit of length is meters and the magnitude of the force is measured in newtons, then the work done is 36 J.

Problem Set 8.1

Ref: Calculus- James Stewart- 8th Edition. Page# 812, 813

1. If $\vec{a} = \langle 6,-2,3\rangle$ and $\vec{b} = \langle 2,5,-1\rangle$, find $\vec{a}\cdot\vec{b}$.
2. If $\vec{a} = 3\hat{\imath} + 2\hat{\jmath} - \hat{k}$ and $\vec{b} = 4\hat{\imath} + 5\hat{k}$, find $\vec{a}\cdot\vec{b}$.
3. Find the angle between the vectors.
   a) $\vec{a} = \langle 1,-4,1\rangle,\ \vec{b} = \langle 0,2,-2\rangle$
   b) $\vec{a} = \langle -1,3,4\rangle,\ \vec{b} = \langle 5,2,1\rangle$
   c) $\vec{a} = 4\hat{\imath} - 3\hat{\jmath} + \hat{k},\ \vec{b} = 2\hat{\imath} - \hat{k}$
   d) $\vec{a} = 8\hat{\imath} - \hat{\jmath} + 4\hat{k},\ \vec{b} = 4\hat{\jmath} + 2\hat{k}$
4. Determine whether the given vectors are orthogonal, parallel, or neither.
   a) $\vec{a} = -8\hat{\imath} + 12\hat{\jmath} + 4\hat{k},\ \vec{b} = 6\hat{\imath} - 9\hat{\jmath} - 3\hat{k}$
   b) $\vec{a} = \langle 4,5,-2\rangle,\ \vec{b} = \langle 3,-1,5\rangle$
   c) $\vec{a} = 3\hat{\imath} - \hat{\jmath} + 3\hat{k},\ \vec{b} = 5\hat{\imath} + 9\hat{\jmath} - 2\hat{k}$

d) $\vec{u} = \langle -5,4,-2\rangle,\ \vec{v} = \langle 3,4,-1\rangle$

e) $\vec{u} = 9\hat{\imath} - 6\hat{\jmath} + 3\hat{k}, \vec{v} = -6\hat{\imath} + 4\hat{\jmath} - 2\hat{k}$

f) $\vec{u} = \langle c,c,c\rangle,\ \vec{v} = \langle c,0,-c\rangle$

5. Find the direction cosines and direction angles of the vector.

   a) $\langle c,0,-c\rangle$ b) $\langle 6,3,-2\rangle$ c) $\hat{\imath} - 2\hat{\jmath} - 3\hat{k}$ d) $\frac{1}{2}\hat{\imath} + \hat{\jmath} + \hat{k}$ e) $\langle c,c,c\rangle$ where $c > 0$

6. Find the scalar projection of $\vec{a}$ onto $\vec{b}$.

   a) $\vec{a} = \langle -1,4,8\rangle, \vec{b} = \langle 12,1,2\rangle$

   b) $\vec{a} = \hat{\imath} + 2\hat{\jmath} + 3\hat{k},\ \vec{b} = 5\hat{\imath} - \hat{k}$

7. Find the scalar projection of $\vec{b}$ onto $\vec{a}$.

   a) $\vec{a} = \langle -5,12\rangle, \vec{b} = \langle 4,6\rangle$

   b) $\vec{a} = \langle 4,7,-4\rangle, \vec{b} = \langle 3-1,1\rangle$

   c) $\vec{a} = 3\hat{\imath} - \hat{\jmath} + 3\hat{k},\ \vec{b} = 5\hat{\imath} + 9\hat{\jmath} - 2\hat{k}$

8. Find the work done by a force $\vec{F} = 8\hat{\imath} - 6\hat{\jmath} + 9\hat{k}$ that moves an object from the point $(0,10,8)$ to the point $(6,12,20)$ along a straight line. The distance is measured in meters and the force in newtons.
9. A wagon is pulled horizontally by exerting a constant force of 10 lb on the handle at an angle of $60°$ with the horizontal. How much work is done in moving the wagon 50 ft?
10. A force $\vec{F} = 3\hat{\imath} - \hat{\jmath} + 2\hat{k}$ lb is applied to a point that moves on a line from $P(-1,1,2)$ to $Q(3,0,-2)$. If distance is measured in feet, how much work in done?

The Cross Product

If $\vec{a} = \langle a_1, a_2, a_3\rangle$ and $\vec{b} = \langle b_1, b_2, b_3\rangle$, then the cross product of $\vec{a}$ and $\vec{b}$ is the vector

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

- The cross product $\vec{a} \times \vec{b}$ of two vectors $\vec{a}$ and $\vec{b}$ is a vector. For this reason, it is also called **vector product**.
- Note that, $\vec{a} \times \vec{b}$ is only defined when $\vec{a}$ and $\vec{b}$ are three-dimensional vectors.
- The vector $\vec{a} \times \vec{b}$ is orthogonal to both $\vec{a}$ and $\vec{b}$.
- If θ is the angle between the vectors $\vec{a}$ and $\vec{b}$, then the **magnitude** of $\vec{a} \times \vec{b}$ is,

$$\left|\vec{a} \times \vec{b}\right| = |\vec{a}|\left|\vec{b}\right| \sin\theta$$

- Two nonzero vectors $\vec{a}$ and $\vec{b}$ are **parallel** if and only if

$$\vec{a} \times \vec{b} = \vec{0}$$

Area of the Paralellogram: If $\vec{a}$ and $\vec{b}$ are adjacent sides, then the area of the parallelogram is defined as the magnitude of the cross product $\vec{a} \times \vec{b}$, that is,

$$A = |\vec{a} \times \vec{b}|$$

Area of the Triangle: Half the area of the parallelogram, that is,

$$A = \frac{1}{2}|\vec{a} \times \vec{b}|$$

Algebraic Properties: If $\vec{a}, \vec{b}, and\ \vec{c}$ are any vectors in 3-space and k is any scalar, then

1. $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$
2. $\vec{a} \times (\vec{b} + \vec{c}) = (\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c})$
3. $(\vec{a} + \vec{b}) \times \vec{c} = (\vec{a} \times \vec{c}) + (\vec{b} \times \vec{c})$
4. $k(\vec{a} \times \vec{b}) = (k\vec{a}) \times \vec{b} = \vec{a} \times (k\vec{b})$
5. $\vec{a} \times \vec{0} = \vec{0} \times \vec{a} = \vec{0}$
6. $\vec{a} \times \vec{a} = \vec{0}$

Some frequently used cross products,

$$\hat{\imath} \times \hat{\jmath} = \hat{k} \qquad \hat{\jmath} \times \hat{k} = \hat{\imath} \qquad \hat{k} \times \hat{\imath} = \hat{\jmath}$$

$$\hat{\jmath} \times \hat{\imath} = -\hat{k} \qquad \hat{k} \times \hat{\jmath} = -\hat{\imath} \qquad \hat{\imath} \times \hat{k} = -\hat{\jmath}$$

The results are easy to obtain. For example,

$$\hat{\imath} \times \hat{\jmath} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} \hat{\imath} - \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} \hat{\jmath} + \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \hat{k} = \hat{k}$$

On the other hand,

$$\hat{\imath} \times \hat{\imath} = \vec{0} \qquad \hat{\jmath} \times \hat{\jmath} = \vec{0} \qquad \hat{k} \times \hat{k} = \vec{0}$$

Scalar Triple Product: Let, $\vec{a} = \langle a_1, a_2, a_3 \rangle, \vec{b} = \langle b_1, b_2, b_3 \rangle, \vec{c} = \langle c_1, c_2, c_3 \rangle$. The product $\vec{a} \cdot (\vec{b} \times \vec{c})$ or, $\vec{b} \cdot (\vec{c} \times \vec{a})$ or, $\vec{c} \cdot (\vec{a} \times \vec{b})$ is called the scalar triple product of the vectors $\vec{a}, \vec{b}, \text{and } \vec{c}$. We can write scalar triple product as a determinant:

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

- The **volume of the parallelepiped** determined by the vectors $\vec{a}, \vec{b}, \text{and } \vec{c}$ is the magnitude of their scalar triple product:

$$V = |\vec{a} \cdot (\vec{b} \times \vec{c})|$$

- If the volume of the parallelepiped is 0, then the vectors must lie in the same plane, that is, they are **coplanar**. The vectors $\vec{a}, \vec{b}, \text{and } \vec{c}$ are coplanar if

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = 0$$

Example: Find a vector that is orthogonal to both of the vectors $\vec{u} = \langle 2, -1, 3 \rangle$ and $\vec{v} = \langle -7, 2, -1 \rangle$.

Solution: The vector $\vec{u} \times \vec{v}$ is orthogonal to both $\vec{u}$ and $\vec{v}$. We compute that

$$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & -1 & 3 \\ -7 & 2 & -1 \end{vmatrix}$$
$$= (1-6)\hat{\imath} - (-2+21)\hat{\jmath} + (4-7)\hat{k}$$
$$= -5\hat{\imath} - 19\hat{\jmath} - 3\hat{k}.$$

Example: Find the area of the triangle with vertices $P(1,4,6), Q(-2,5,-1)$, and $R(1,-1,1)$.

Solution: At first we should find the adjacent sides.

Here, $\overrightarrow{PQ} = Q - P = (-2-1)\hat{\imath} + (5-4)\hat{\jmath} + (-1-6)\hat{k} = -3\hat{\imath} + \hat{\jmath} - 7\hat{k}$

$\overrightarrow{PR} = Q - P = (1-1)\hat{\imath} + (-1-4)\hat{\jmath} + (1-6)\hat{k} = -5\hat{\jmath} - 5\hat{k}$

So, $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ are adjacent sides.

Now, $\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ -3 & 1 & -7 \\ 0 & -5 & -5 \end{vmatrix}$

$$= (-5-35)\hat{\imath} - (15-0)\hat{\jmath} + (15-0)\hat{k} = -40\hat{\imath} - 15\hat{\jmath} + 15\hat{k}$$

Therefore, the area of the triangle is

$$A = \frac{1}{2}\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right| = \frac{1}{2}\sqrt{(-40)^2 + (-15)^2 + (15)^2} = \frac{5}{2}\sqrt{82}.$$

Example: Use scalar triple product to show that the vectors $\vec{a} = \hat{\imath} + 4\hat{\jmath} - 7\hat{k}$, $\vec{b} = 2\hat{\imath} - \hat{\jmath} + 4\hat{k}$, and $\vec{c} = -9\hat{\jmath} + 18\hat{k}$ are coplanar.

Solution: $\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} 1 & 4 & -7 \\ 2 & -1 & 4 \\ 0 & -9 & 18 \end{vmatrix}$

$$= 1(-18+36) - 4(36-0) - 7(-18-0) = 0$$

Since the scalar triple product of the vectors is 0, they are coplanar.

Problem Set 8.2 (Cross Product)

Ref: Calculus- James Stewart- 8th Edition. Page# 821

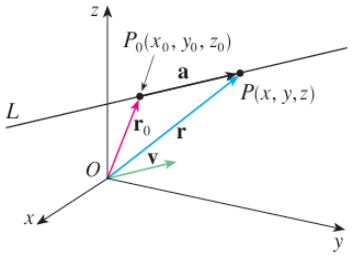
1. Show that $\vec{u} \times \vec{u} = 0$ for any vector $\vec{u}$ in 3-space.
2. Let $\vec{u} = \langle 2,-1,3\rangle$, $\vec{v} = \langle 0,1,7\rangle$, and $\vec{w} = \langle 1,4,5\rangle$. Find
   a) $\vec{u} \times (\vec{v} \times \vec{w})$
   b) $(\vec{u} \times \vec{v}) \times \vec{w}$
3. Find two unit vectors orthogonal to both $\langle 3,2,1\rangle$ and $\langle -1,1,0\rangle$.
4. Find two unit vectors orthogonal to both $\hat{\jmath} - \hat{k}$ and $\hat{\imath} + \hat{\jmath}$.

5. Find the area of the parallelogram with vertices $A(-3,0), B(-1,3), C(5,2)$, and $D(3,-1)$.
6. Find the area of the parallelogram with vertices $P(1,0,2), Q(3,3,3), R(7,5,8)$, and $S(5,2,7)$.
7. Find the volume of the parallelepiped determined by the vectors $\vec{a}, \vec{b}$, and $\vec{c}$ where $\vec{a} = (1,2,3)$, $\vec{b} = \langle -1,1,2\rangle, \vec{c} = \langle 2,1,4\rangle$.
8. Find the volume of the parallelepiped determined by the vectors $\vec{a}, \vec{b}$, and $\vec{c}$ where $\vec{a} = \hat{\imath} + \hat{\jmath}, \vec{b} = \hat{\jmath} + \hat{k}, \vec{c} = \hat{\imath} + \hat{\jmath} + \hat{k}$.
9. Use the scaler triple product to verify that the vectors $\vec{u} = \hat{\imath} + 5\hat{\jmath} - 2\hat{k}, \hat{v} = 3\hat{\imath} - \hat{\jmath}$, and $\vec{w} = 5\hat{\imath} + 9\hat{\jmath} - 4\hat{k}$ are coplanar.
10. Find the area of the triangle that is determined by the points $P(2,2,0), Q(-1,0,2)$, and $R(0,4,3)$.
11. Find the area of the triangle with vertices $P(1,5,-2), Q(0,0,0)$, and $R(3,5,1)$.

Chapter 9

Equations of Lines and Planes

Lines



A line L in three-dimensional space is determined when we know a point $P_0(x_0, y_0, z_0)$ on L and the direction of L. In three dimensions the direction of a line is conveniently described by a vector, so we let v be a vector parallel to L. Let $P(x, y, z)$ be an arbitrary point on L and let $\overrightarrow{r_0}$ and $\vec{r}$ be the position vectors of P_0 and P. If $\vec{a}$ is the vector with representation $\overrightarrow{P_0P}$, as in Figure, then the Triangle Law for vector addition gives $r = r_0 + \vec{a}$. But, since $\vec{a}$ and $\vec{v}$ are parallel vectors, there is a scalar t such that $\vec{a} = t\vec{v}$. Thus

$$\vec{r} = \overrightarrow{r_0} + t\vec{v}$$

which is a **vector equation** of L. Each value of the parameter t gives the position vector $\vec{r}$ of a point on L. If the vector $\vec{v}$ that gives the direction of the line L is written in component form as $\vec{v} = \langle a, b, c \rangle$, then we have $t\vec{v} = \langle ta, tb, tc \rangle$. We can also write $\vec{r} = \langle x, y, z \rangle$ and $\overrightarrow{r_0} = \langle x_0, y_0, z_0 \rangle$, so the vector equation becomes

$$\langle x, y, z \rangle = \langle x_0 + at, y_0 + bt, z_0 + ct \rangle$$

Two vectors are equal if and only if corresponding components are equal. Therefore we have the three scalar equations:

$$x = x_0 + at \qquad y = y_0 + bt \qquad z = z_0 + ct$$

where $t \in \boldsymbol{R}$. These equations are called parametric equations of the line L through the point $P_0(x_0, y_0, z_0)$ and parallel to the vector $\vec{v} = \langle a, b, c \rangle$. Each value of the parameter t gives a point (x, y, z) on L.

Another way of describing a line L is to eliminate the parameter t. We can solve each of these equations for t:

$$t = \frac{x - x_0}{a} \qquad t = \frac{y - y_0}{b} \qquad t = \frac{z - z_0}{c}$$

Equating the results, we obtain

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

These equations are called **symmetric equations** of L.

Vector Equation of line: $\vec{r} = \overrightarrow{r_0} + t\vec{v}$

Parametric Equations of line: $x = x_0 + at \qquad y = y_0 + bt \qquad z = z_0 + ct$

Symmetric equations of line: $\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$

Example: Find the vector equation, parametric equations and symmetric equations for the line through the point $(5,1,3)$ and parallel to the vector $\hat{\imath} + 4\hat{\jmath} - 2\hat{k}$.

Solution: Here, $r_0 = \langle 5,1,3\rangle = 5\hat{\imath} + \hat{\jmath} + 3\hat{k}$ and $\vec{v} = \hat{\imath} + 4\hat{\jmath} - 2\hat{k}$.

Vector equation of line is,

$$\vec{r} = \overrightarrow{r_0} + t\vec{v}$$
$$= (5\hat{\imath} + \hat{\jmath} + 3\hat{k}) + t(\hat{\imath} + 4\hat{\jmath} - 2\hat{k})$$
$$= (5 + t)\hat{\imath} + (1 + 4t)\hat{\jmath} + (3 - 2t)\hat{k}$$

Parametric equations of line are,

$$x = 5 + t \quad y = 1 + 4t \quad z = 3 - 2t$$

From the parametric equations of line, we can compute t,

$$t = x - 5 \qquad t = \frac{y-1}{4} \qquad t = \frac{z-3}{2}$$

Therefore, symmetric equations of line are,

$$\frac{x-5}{1} = \frac{y-1}{4} = \frac{z-3}{2}.$$

Example: Find the vector equation, parametric equations and symmetric equations of the line that passes through the points $A(2,4,-3)$ and $B(3,-1,1)$.

Solution: The vector $\vec{v}$ with representation $\overrightarrow{AB}$ is parallel to the line and

$$\vec{v} = \overrightarrow{AB} = B - A$$
$$= \langle 3 - 2, -1 - 4, 1 + 3\rangle = \langle 1, -5, 4\rangle$$

Taking the point $A(2,4,-3)$ as $\overrightarrow{r_0}$, the vector equation of line is,

$$\vec{r} = \overrightarrow{r_0} + t\vec{v}$$
$$= \langle 2,4,-3\rangle + t\langle 1,-5,4\rangle$$
$$= \langle 2 + t, 4 - 5t, -3 + 4t\rangle$$

Parametric equations are

$$x = 2 + t \quad y = 5 - 5t \quad z = -3 + 4t$$

and symmetric equations are

$$\frac{x-2}{1} = \frac{y-5}{-5} = \frac{z+3}{4}.$$

Example: Show that the lines L_1 and L_2 are skew lines (they do not intersect and are not parallel) with parametric equations

$$L_1: \quad x = 1 + t, \quad y = -2 + 3t, \quad z = 4 - t$$
$$L_2: \quad x = 2s, \qquad y = 3 + s, \qquad z = -3 + 4s.$$

Solution: Let, direction vector of L_1 is $\overrightarrow{v_1} = \langle a_1, a_2, a_3\rangle = \langle 1,3,-1\rangle$, direction vector of L_2 is $\overrightarrow{v_2} = \langle b_1, b_2, b_3\rangle = \langle 2,1,4\rangle$.

Now, $\frac{a_1}{b_1} = \frac{1}{2}$ $\frac{a_2}{b_2} = \frac{3}{1}$ $\frac{a_3}{b_3} = -\frac{1}{4}$

Since $\frac{a_1}{b_1} \neq \frac{a_2}{b_2} \neq \frac{a_3}{b_3}$, the lines are not parallel.

If L_1 and L_2 had a point of intersection, then we can write

$1 + t = 2s$ (i)

$-2 + 3t = 3 + s$ (ii)

$4 - t = -3 + 4s$ (iii)

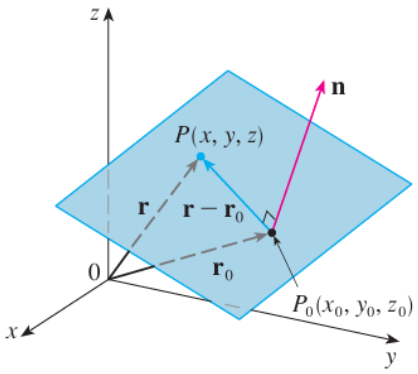
Solving the first two equations we have,

$t = \frac{11}{5}$ and $s = \frac{8}{5}$

Putting these values in equation (iii), we get $\frac{9}{5}$ in the left side and $\frac{17}{5}$ in the right side. That means the values of t and s do not satisfy the third equation. Therefore, there are no values of t and s that satisfy the three equations, so L_1 and L_2 do not intersect. Thus L_1 and L_2 are skew lines.

Planes

Although a line in space is determined by a point and a direction, a plane in space is more difficult to describe. A single vector parallel to a plane is not enough to convey the "direction" of the plane, but a vector perpendicular to the plane does completely specify its direction. Thus a plane in space is determined by a point $P_0(x_0, y_0, z_0)$ in the plane and a vector $\vec{n}$ that is orthogonal to the plane. This orthogonal vector $\vec{n}$ is called a **normal vector**. Let $P(x, y, z)$ be an arbitrary point in the plane, and let $\overrightarrow{r_0}$ and $\vec{r}$ be the position vectors of P_0 and P. Then the vector $\vec{r} - \overrightarrow{r_0}$ is represented by $\overrightarrow{P_0P}$. The normal vector $\vec{n}$ is orthogonal to every vector in the given plane. In particular, $\vec{n}$ is orthogonal to $\vec{r} - \overrightarrow{r_0}$ and so we have



$$\vec{n} \cdot (\vec{r} - \overrightarrow{r_0}) = 0$$

This equation is called a vector equation of line.

To obtain a scalar equation for the plane, we write $\vec{n} = \langle a, b, c\rangle$, $\vec{r} = \langle x, y, z\rangle$, and $\overrightarrow{r_0} = \langle x_0, y_0, z_0\rangle$. Then the vector equation becomes

$$\langle a, b, c\rangle \cdot \langle x - x_0, y - y_0, z - z_0\rangle = 0$$

or, $a(x-x_0)+b(y-y_0)+c(z-z_0)=0$

We can rewrite the equation of a plane as

$$ax+by+cz+d=0$$

where $d=-(ax_0+by_0+cz_0)$.

- Two planes are **parallel** if their normal vectors are parallel.
- Angle between two planes, $\cos\theta=\frac{\overrightarrow{n_1}\cdot\overrightarrow{n_2}}{|\overrightarrow{n_1}||\overrightarrow{n_2}|}$

Example: Find an equation of the plane through the point $(2,4,-1)$ with normal vector $\hat{n}=\langle 2,3,4\rangle$.

Solution: The equation of plane is

$$\vec{n}\cdot(\vec{r}-\overrightarrow{r_0})=0$$

or, $\langle 2,3,4\rangle\cdot\langle x-2,y-4,z+1\rangle=0$

or, $2(x-2)+3(y-4)+4(z+1)=0$

or, $2x-4+3y-12+4z+4=0$

or, $2x+3y+4z=12$.

Example: Find an equation of the plane through the points $P(1,3,2)$, $Q(3,-1,6)$, and $R(5,2,0)$.

Solution: The vectors $\vec{a}$ and $\vec{b}$ corresponding to $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ are

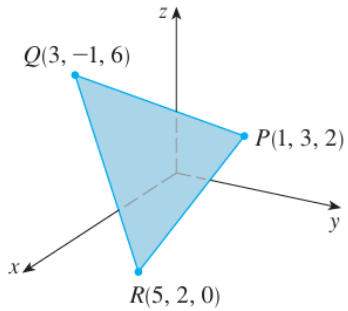
$\vec{a}=\langle 2,-4,4\rangle$ and $\vec{b}=\langle 4,-1,-2\rangle$.

Since both $\vec{a}$ and $\vec{b}$ lie in the plane, their cross product $\vec{a}\times\vec{b}$ is orthogonal to the plane and can be taken as the normal vector. Thus

$$\vec{n}=\vec{a}\times\vec{b}=\begin{vmatrix}\hat{\imath}&\hat{\jmath}&\hat{k}\\2&-4&4\\4&-1&-2\end{vmatrix}$$

$$=(8+4)\hat{\imath}-(-4-16)\hat{\jmath}+(-2+16)\hat{k}=12\hat{\imath}+20\hat{\jmath}+14\hat{k}$$

With the point $P(1,3,2)$ and the normal vector $\vec{n}$, an equation of the plane is



$$\vec{n}\cdot(\vec{r}-\overrightarrow{r_0})=0$$

or, $\langle 12,20,14\rangle\cdot\langle x-1,y-3,z-2\rangle=0$

or, $12(x-1)+20(y-3)+14(z-2)=0$

or, $12x+20y+14z-100=0$

or, $6x+10y+7z=50$.

Example: Find the angle between the planes $x+y+z=1$ and $x-2y+3z=1$.

Solution: The normal vectors of these planes are $\overrightarrow{n_1}=\langle 1,1,1\rangle$ and $\overrightarrow{n_2}=\langle 1,-2,3\rangle$. If θ is the angle between the planes, then

$$\cos\theta = \frac{\overrightarrow{n_1}\cdot\overrightarrow{n_2}}{|\overrightarrow{n_1}||\overrightarrow{n_2}|}$$

$$= \frac{1\cdot 1 + 1(-2) + 1\cdot 3}{\sqrt{1^2+1^2+1^2}\sqrt{1^2+(-2)^2+3^2}} = \frac{2}{\sqrt{42}}$$

Therefore, $\theta = \cos^{-1}\left(\frac{2}{\sqrt{42}}\right) \approx 72°$.

Problem Set 9

Ref: Calculus- James Stewart- 8th Edition. Page# 831, 832.

1. Find the vector equation, parametric equations and symmetric equations for the line through the point $(6,-5,2)$ and parallel to the vector $\langle 1,3,-\frac{2}{3}\rangle$.
2. Find the vector equation, parametric equations and symmetric equations for the line through the point $(0,14,-10)$ and parallel to the line $x=-1+2t, y=6-3t$, and $z=3+9t$.
3. Find the parametric equations and symmetric equations for the line through the points $(0,\frac{1}{2},1)$ and $(2,1,-3)$.
4. Find the parametric equations and symmetric equations for the line through $(2,1,0)$ and perpendicular to both $\hat{\imath}+\hat{\jmath}$ and $\hat{\jmath}+\hat{k}$.
5. Find the parametric equations and symmetric equations for the line through $(-6,2,3)$ and parallel to the line $\frac{1}{2}x=\frac{1}{3}y=z+1$.
6. Determine whether the lines L_1 and L_2 are parallel, skew, or intersecting. If they intersect, find the point of intersection.
   a) L_1: $x=3+2t,\quad y=4-t,\quad z=1+3t$
      L_2: $x=1+4s,\quad y=3-2s,\quad z=4+5s$
   b) L_1: $x=5-12t,\quad y=3+9t,\quad z=1-3t$
      L_2: $x=3+8s,\quad y=-6s,\quad z=7+2s$
   c) L_1: $\frac{x-2}{1}=\frac{y-3}{-2}=\frac{z-1}{-3}$
      L_2: $\frac{x-3}{1}=\frac{y+4}{3}=\frac{z-2}{-7}$
   d) L_1: $\frac{x}{1}=\frac{y-1}{-1}=\frac{z-2}{3}$
      L_2: $\frac{x-2}{2}=\frac{y-3}{-2}=\frac{z}{7}$
7. Find an equation of the plane through the origin and perpendicular to the vector $\langle 1,-2,5\rangle$.
8. Find an equation of the plane through the point $(5,3,5)$ and with normal vector $2\hat{\imath}+\hat{\jmath}-\hat{k}$.
9. Find an equation of the plane through the point $(2,0,1)$ and perpendicular to the line $x=3t, y=2-t, z=3+4t$.
10. Find an equation of the plane through the point $(1,-1,-1)$ and parallel to the plane $5x-y-z=6$.
11. Find an equation of the plane through the points $(2,1,2), (3,-8,6)$, and $(-2,-3,1)$.

12. Find an equation of the plane through the points $(3,0,-1)$, $(-2,-2,3)$, and $(7,1,-4)$.
13. Determine whether the planes are parallel, perpendicular, or neither. If neither, find the angle between them.
    a) $x + 4y - 3z = 1,\ -3x + 6y + 7z = 0$
    b) $x + 2y - z = 2,\ 2x - 2y + z = 1$
    c) $2x - 3y = z,\ 4x = 3 + 6y + 2z.$